

Task 4: Optimal Externalization Threshold

Method

In Task 4, the aim is to decide when the LP should externalize a trade instead of keeping it internally.

The Task 3 model gives the predicted probability that a trade will become adverse for the LP. I used this predicted adversity probability to decide whether to externalize or internalize a trade.

The rule used was:

If predicted adversity probability is greater than the threshold θ , the trade is externalized.

If predicted adversity probability is less than or equal to θ , the trade is internalized.

For different values of θ , I calculated validation PnL and selected the threshold that gave the best validation PnL. This was done for each client and each horizon $\tau=5,10,15,20,25,30$.

Theta_star is the best threshold found on validation data, and final_pnl is the PnL on the test data.

Validation PnL Plot

The PnL validation plot shows how PnL changes as the threshold changes.

When the threshold is very low, many trades are externalized. This protects the LP from bad trades, but it also removes many good trades that could have made profit.

When the threshold is very high, very few trades are externalized. This means the LP keeps more trades internally, but also keeps more adverse trades and can lose money.

The best threshold is somewhere in between. It tries to keep profitable trades while externalizing trades that are likely to be adverse.

Over-Externalization vs Under-Externalization

Over-externalization happens when the threshold is too low. In this case, the LP externalizes too many trades. Some of these trades may have actually been profitable, so the LP loses possible profit.

Under-externalization happens when the threshold is too high. In this case, the LP keeps too many trades internally. Some of these trades may be adverse, so the LP faces losses.

So the main trade-off is between avoiding bad trades and not giving up too many good trades.

Global vs Client-Specific Thresholds

The plot shows the general global behaviour of validation PnL across horizons.

However, the submitted CSV uses client-specific thresholds. I used client-specific thresholds because all clients do not behave the same way.

Some clients are safer and more profitable for the LP, while some clients are more adverse, especially at longer horizons. If one single global threshold is used for all clients, it may be too strict for good clients and too loose for risky clients.

So client-specific thresholds are more suitable because they adjust to each client's trading behaviour.

Final Test PnL

The final test PnL is reported in `task4_results.csv`.

Clients A, B, and C generally give stronger and more positive results. Clients D, E, and F are more difficult, especially at longer horizons, because their trades are more adverse for the LP.

Some client-horizon pairs may still have negative test PnL. This does not necessarily mean the implementation is wrong. It can happen because the threshold was selected on validation data, while final PnL is tested on unseen test data.

Overall, the strategy does not externalize every trade. It selectively externalizes trades only when the predicted adversity probability is high enough. This is better than blindly externalizing all trades or keeping all trades internally.